

A Comparative Benchmark of VGG16, ResNet50, and EfficientNet-B0 on CIFAR-10

Aviral Singh

Abstract—This work benchmarks three widely used convolutional neural network (CNN) architectures — VGG16, ResNet50, and EfficientNet-B0 — on the CIFAR-10 image classification task using ImageNet-pretrained weights and full fine-tuning. The comparison covers parameter count, training time, inference latency, test accuracy, and per-class error structure, and includes Grad-CAM visualizations for qualitative explainability. Results show that EfficientNet-B0 achieves 95.88% test accuracy with only 4.02M parameters and 0.61 ms/image inference, ResNet50 attains the highest accuracy (96.51%) at moderate cost, and VGG16 is strictly Pareto-dominated. The findings illustrate the practical impact of architectural advances such as residual connections and compound scaling.

I. INTRODUCTION

Deep convolutional neural networks have driven much of the progress in computer vision over the past decade. Successive architectures have improved not only accuracy but also parameter efficiency and inference speed. This report benchmarks three representative architectures from different generations of CNN design: VGG16 [1], ResNet50 [2], and EfficientNet-B0 [3]. All three are fine-tuned from ImageNet-pretrained weights on CIFAR-10 [5], and compared on accuracy, parameter count, training time, inference latency, confusion structure, and Grad-CAM-based [4] interpretability. The goal is to characterize the trade-offs each architecture offers under a consistent training protocol.

II. MODELS AND METHODOLOGY

A. Dataset

CIFAR-10 consists of 60,000 color images of size 32×32 across ten classes, partitioned by the dataset providers into 50,000 training and 10,000 test images. The training set is further split, stratified by class, into 45,000 training and 5,000 validation images. Images are resized to 128×128 , normalized with ImageNet statistics, and augmented during training with random horizontal flipping and random cropping with padding.

B. Architectures

VGG16 [1] is a 16-layer network consisting of thirteen 3×3 convolutional layers arranged in five blocks separated by max-pooling, followed by three fully connected layers (Fig. 1). The use of small, stacked filters approximates larger receptive fields with fewer parameters per layer, but the two fully connected layers of 4,096 units dominate the total parameter count (~ 134 M).

ResNet50 [2] is a 50-layer residual network built from *bottleneck* blocks ($1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$ convolutions) grouped into four stages. Each block adds a skip connection from its input to its output (Fig. 2), which mitigates the vanishing-gradient problem in deep networks and allows much deeper models to be trained effectively. ResNet50 has approximately 25M parameters.

EfficientNet-B0 [3] is the baseline model of the EfficientNet family. It is composed of Mobile Inverted Bottleneck (MBConv) blocks that use depthwise-separable convolutions and squeeze-and-excitation modules. The architecture was discovered via neural architecture search and is designed to balance depth, width, and resolution via compound scaling. EfficientNet-B0 contains only about 5M parameters yet matches or exceeds the accuracy of much larger models on ImageNet.

C. Training Setup

All three models are initialized from ImageNet-pretrained weights, with the final classification layer replaced by a new linear layer for ten classes. The full network is fine-tuned (no layers frozen) using the Adam optimizer with learning rate 1×10^{-4} , batch size 64, and cross-entropy loss. Training runs for up to 15 epochs with early stopping (patience 4 epochs) monitoring validation loss; the best-validation checkpoint is restored. Mixed-precision training is used to accelerate computation on the GPU.

III. RESULTS

A. Training Dynamics

Figure 3 shows the loss and accuracy curves for each model. VGG16 ran for ten epochs before early stopping; its training loss continued to decrease while validation loss began to rise after epoch 6, indicating overfitting. ResNet50 converged remarkably fast — validation accuracy reached approximately 96% by epoch 2 — and was stopped at epoch 7. EfficientNet-B0 produced the smoothest curves: at epoch 1, validation accuracy exceeds training accuracy, which is the expected signature of strong built-in regularization (dropout and stochastic depth applied only during training). Its validation loss flattens rather than rising, indicating no significant overfitting.

B. Test Performance and Confusion Structure

Table 1 summarizes parameter count, training time, inference latency, and test accuracy for each model. ResNet50 obtains the highest test accuracy at 96.51%, followed closely by EfficientNet-B0 at 95.88%, with VGG16 at 93.55%. Since the test set is balanced (1,000 images per class), macro-averaged precision, recall, and F1-score closely track accuracy for all three models.

Table 1. Benchmark Summary on the CIFAR-10 Test Set

Model	Params (M)	Train (s)	Infer. (ms/img)	Acc. (%)
VGG16	134.30	1488	1.847	93.55
ResNet50	23.53	598	0.905	96.51
EfficientNet-B0	4.02	905	0.610	95.88

The confusion matrices in Figure 4 reveal that all three models fail on essentially the same visually ambiguous class pairs. The cat $\leftrightarrow$ dog confusion is the single dominant error mode in every model, followed by vehicle confusions (airplane $\leftrightarrow$ ship,



Figure 1. VGG16 architecture: five blocks of stacked 3×3 convolutions with increasing channel depth, separated by 2×2 max-pooling, followed by three fully connected layers. The spatial size halves at each pooling stage. Adapted from [1].

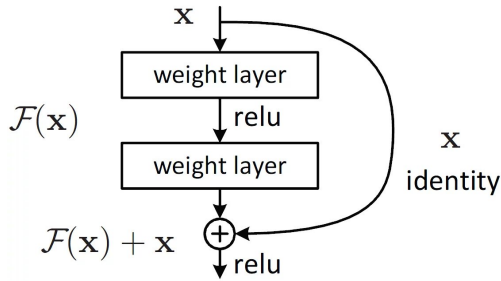


Figure 2. Residual block used in ResNet. The block computes $\mathcal{F}(x)$ and adds the identity-mapped input x , producing $\mathcal{F}(x) + x$. Adapted from [2].

automobile $\leftrightarrow$ truck). The consistency of these errors across architectures suggests that the residual difficulty is intrinsic to CIFAR-10 at low resolution rather than a deficiency of any one model. VGG16’s errors are notably more spread across off-diagonal cells, while ResNet50 and EfficientNet-B0 concentrate their mistakes on a small set of hard pairs.

C. Grad-CAM Visualizations

Figure 5 compares Grad-CAM heatmaps from the last convolutional block of each model. ResNet50’s heatmaps are the most tightly localized, with a single concentrated region of activation centered on the object. EfficientNet-B0’s heatmaps are similarly focused but tend to cover a broader portion of the object’s extent. VGG16’s heatmaps are visibly diffuse and occasionally drift onto background regions; on the third example (a ship), VGG16’s attention is concentrated on the dock and waterline rather than the hull, accompanying an incorrect prediction of “automobile.”

A plausible architectural explanation is that VGG16 reduces spatial resolution through five max-pooling stages, so by the last convolutional layer the spatial feature map is small and coarse. ResNet50’s residual connections preserve spatial information through depth, and EfficientNet-B0’s squeeze-and-excitation modules explicitly learn to weight informative regions. Thus, the localization quality of the Grad-CAM heatmaps is itself a reflection of how each architecture treats spatial information.

D. Efficiency Comparison

Figure 6 presents the final comparison across four axes. EfficientNet-B0 is the most parameter-efficient model: with only 4.02M parameters — approximately $33 \times$ fewer than VGG16 and $6 \times$ fewer than ResNet50 — it achieves test accuracy within 0.6 percentage points of ResNet50. It also has the lowest inference latency at 0.61 ms per image. ResNet50 attains the highest test accuracy at moderate parameter cost. VGG16 is dominated across all four metrics: it has the lowest accuracy, the most parameters, the slowest inference, and the longest training time.

A non-obvious observation concerns wall-clock training time. Despite having far fewer parameters than ResNet50, EfficientNet-B0’s training is slower (905 s vs. 598 s). Two factors contribute: EfficientNet ran for nine epochs while ResNet50 was stopped at seven, and EfficientNet’s depthwise-separable convolutions, although FLOP-efficient, are bound by memory bandwidth and therefore underutilize the GPU relative to the dense convolutions in ResNet50. This decoupling between parameter count and training time is an important caveat when interpreting efficiency benchmarks.

IV. DISCUSSION

The combined evidence yields three main observations.

First, the three models trace out a clear Pareto frontier. EfficientNet-B0 and ResNet50 trade small accuracy differences against large efficiency differences, while VGG16 is strictly dominated. This contradicts any expectation of a strict size–accuracy trade-off and demonstrates the cumulative impact of architectural advances such as residual connections and compound scaling.

Second, the error structures of the three models are highly correlated. The shared cat $\leftrightarrow$ dog and vehicle confusions across all three architectures suggest that residual errors stem largely from the intrinsic ambiguity of low-resolution CIFAR-10 images. Further accuracy gains on this dataset likely require higher input resolution or stronger augmentation rather than more architectural depth.

Third, the qualitative Grad-CAM evidence corroborates the quantitative ranking: more accurate models produce more spatially focused explanations, and VGG16’s diffuse heatmaps coincide with its misclassifications. Localization quality and classification quality are not independent.

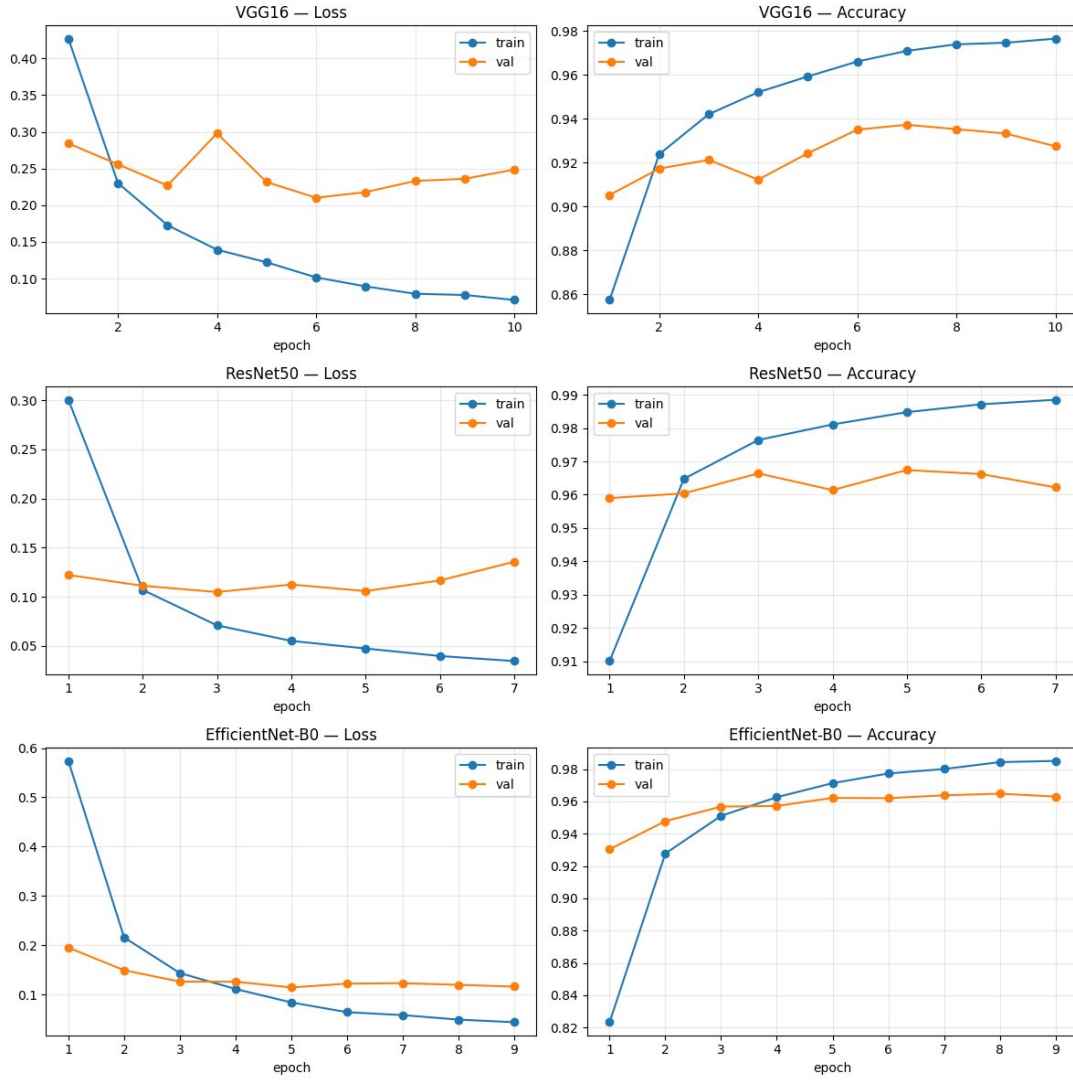


Figure 3. Training and validation curves for each model. VGG16 shows the largest train-validation gap and rising validation loss late in training. ResNet50 converges within two epochs and is stopped early. EfficientNet-B0 has the smallest gap and the most stable validation curve, consistent with its built-in regularization.

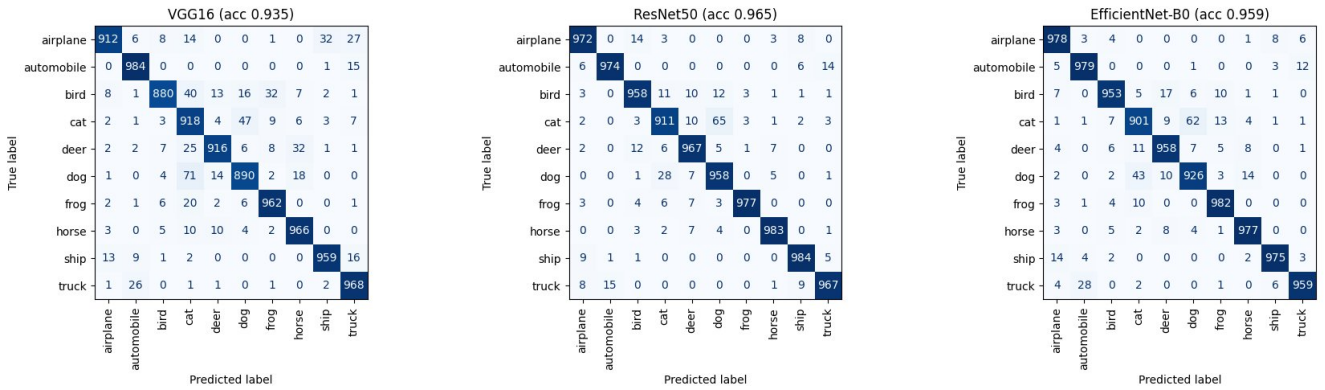


Figure 4. Confusion matrices on the CIFAR-10 test set. Diagonal values represent correct predictions per class (out of 1,000). All three models share similar error patterns, with cat $\leftrightarrow$ dog dominating the off-diagonal entries.

A limitation of this study is that results are reported for a single training run per architecture; the 0.63-point gap between

ResNet50 and EfficientNet-B0 could plausibly lie within seed-to-seed variation. The efficiency comparisons, however, are

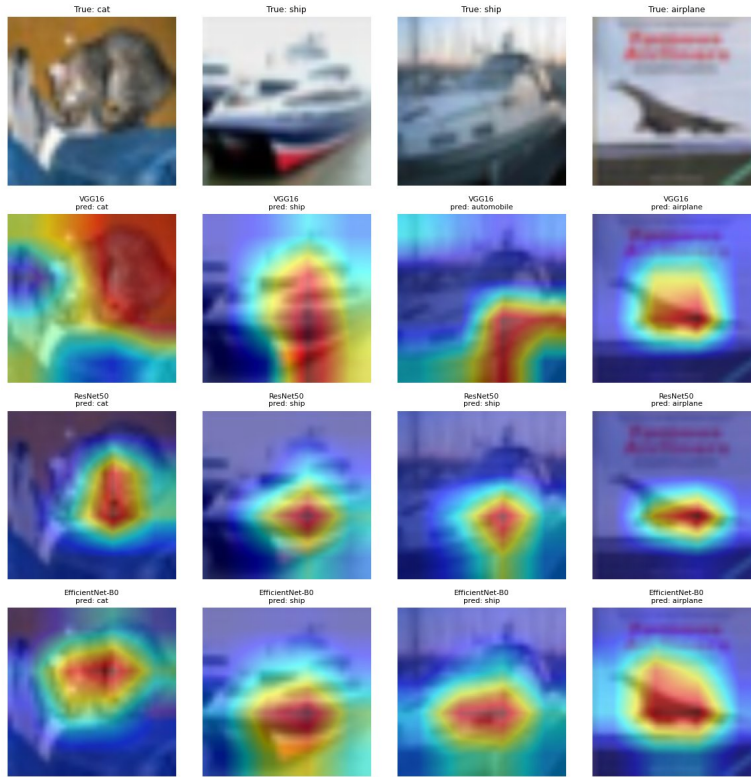


Figure 5. Grad-CAM visualizations on four test images. Top row: original images. Subsequent rows: heatmaps from VGG16, ResNet50, and EfficientNet-B0. ResNet50 produces the most tightly localized attention, EfficientNet-B0 highlights a broader object region, and VGG16 shows diffuse heatmaps. On the third image (ship), VGG16 misclassifies the example as “automobile,” with attention concentrated on the dock rather than the hull.

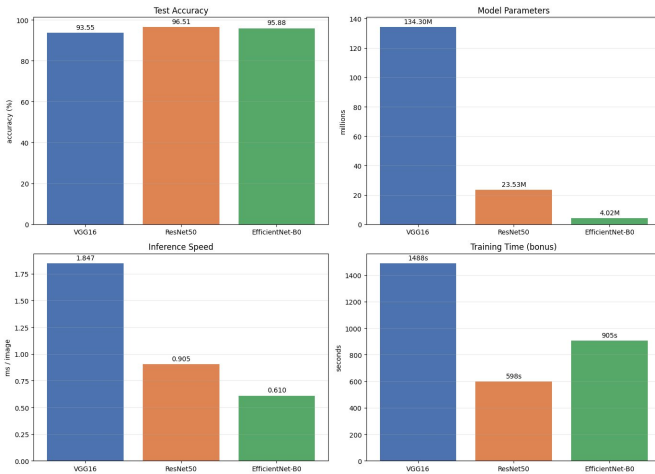


Figure 6. Comparison of test accuracy, parameter count, inference latency, and training time. VGG16 is dominated on every axis; EfficientNet-B0 and ResNet50 occupy the Pareto frontier.

deterministic.

V. CONCLUSION

This report compared VGG16, ResNet50, and EfficientNet-B0 on CIFAR-10 under a unified fine-tuning protocol. ResNet50 achieved the highest accuracy (96.51%), EfficientNet-B0 offered

the best efficiency at near-equal accuracy (95.88% with only 4.02M parameters and 0.61 ms inference), and VGG16 was dominated on every measured axis. Grad-CAM analysis showed that more accurate models also produce more spatially focused explanations, and that the residual errors are concentrated on a small set of intrinsically ambiguous class pairs shared by all three models.

REFERENCES

- [1] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [3] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6105–6114.
- [4] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 618–626.
- [5] A. Krizhevsky, “Learning multiple layers of features from tiny images,” Tech. Rep., Univ. of Toronto, 2009.